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P3 — Electric Circuits

OCR GCSE Physics B (J259) — 21st Century Science · Paper 1 (Breadth in Physics)

P3.1 Current, Charge and Potential Difference

Current (I) is the rate of flow of electric charge. Unit: **ampere (A)**. A current of 1 A means 1 coulomb of charge passes a point each second.

$$\text{Charge (Q)} = \text{current (I)} \times \text{time (t)}$$
$$(\text{C}) = (\text{A}) \times (\text{s})$$

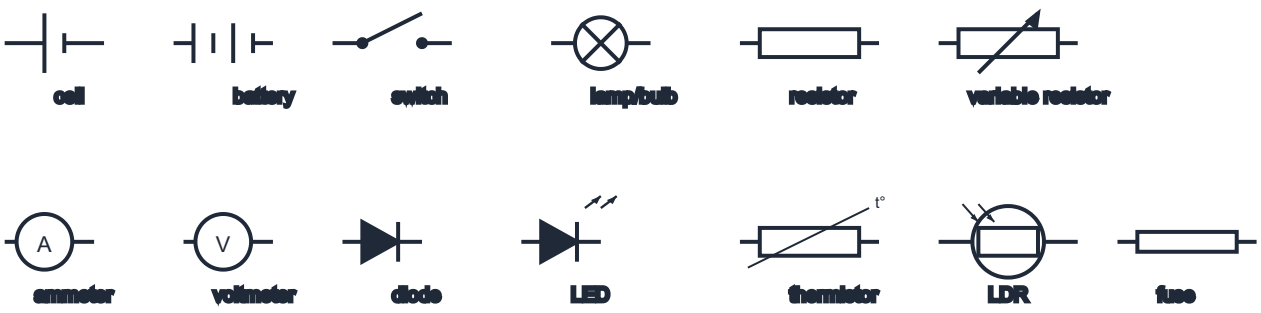
Potential difference (p.d., V) is the energy transferred per coulomb of charge passing between two points. Unit: **volt (V)**. 1 V = 1 J/C.

$$\text{Energy transferred (E)} = \text{charge (Q)} \times \text{potential difference (V)}$$
$$(\text{J}) = (\text{C}) \times (\text{V})$$

Worked Example: A current of 0.5 A flows through a lamp for 2 minutes. How much charge flows?
Q = I × t = 0.5 × 120 = **60 C**

P3.2 Circuit Symbols

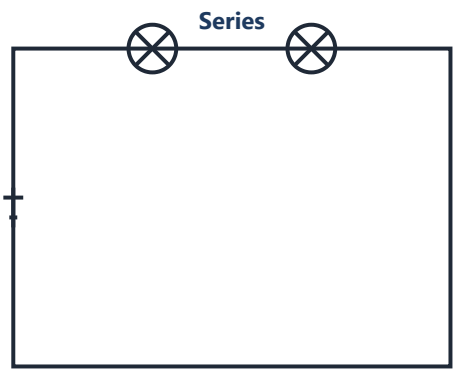
Common Circuit Symbols



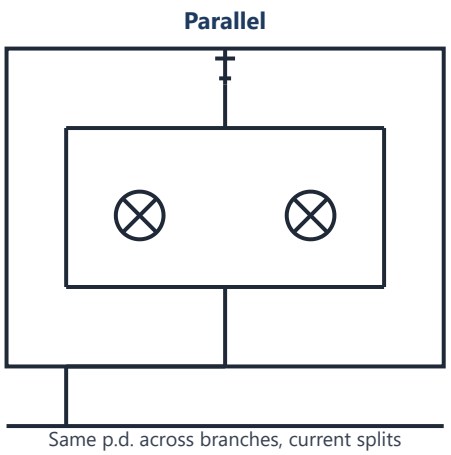
You must recognise and draw these symbols accurately in exams.

P3.3 Series vs Parallel Circuits

Series and Parallel Circuits



Same current, p.d. splits between components



Same p.d. across branches, current splits

	Series	Parallel
Current	Same everywhere: $I = I_1 = I_2$	Splits and recombines: $I_{total} = I_1 + I_2$
P.d.	Splits across components: $V_{total} = V_1 + V_2$	Same across each branch: $V = V_1 = V_2$
Resistance	$R_{total} = R_1 + R_2$	R_{total} is less than either individual resistance

If one breaks	Whole circuit stops	Other branches keep working

P3.4 Resistance & Ohm's Law

Resistance (R) is a measure of how difficult it is for current to flow. Unit: **ohm (Ω)**.

$$V = I \times R \quad \text{(Ohm's Law for ohmic conductors)}$$
$$(V) = (A) \times (\Omega)$$

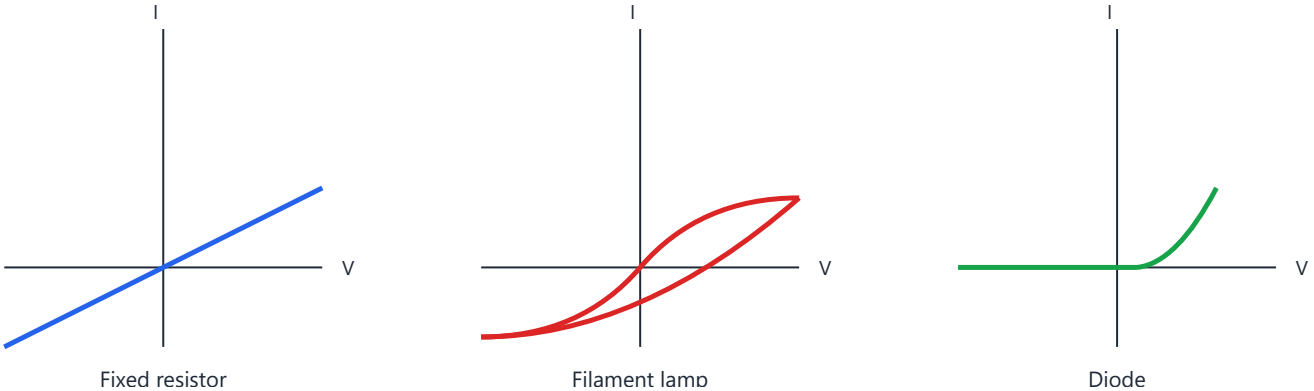
Worked Example: A 12 V battery pushes 0.4 A through a resistor. What is its resistance?
R = V / I = 12 / 0.4 = **30 Ω**

Factors affecting resistance of a wire:

- Longer wire → more resistance (R ∝ length).
- Thicker wire → less resistance (larger cross-sectional area).
- Different material (resistivity).
- Higher temperature → usually more resistance in metals (ions vibrate more).

P3.5 I–V Characteristics

I–V Characteristics



Thermistor and LDR

- **Thermistor**: resistance *decreases* as temperature increases. Used in thermostats and fire alarms.
- **LDR (light-dependent resistor)**: resistance *decreases* as light intensity increases. Used in streetlight sensors and camera exposure.

P3.6 Electrical Power and Energy

Electrical power: $P = V \times I$ or $P = I^2 R$ or $P = V^2 / R$

$(W) = (V) \times (A)$

Energy transferred: $E = P \times t = V \times I \times t$

$(J) = (W) \times (s)$

Worked Example: A kettle is rated 2300 W at 230 V. Calculate the current and the fuse rating (3 A, 5 A or 13 A) it needs.
I = P / V = 2300 / 230 = **10 A**. Nearest fuse above: **13 A**.

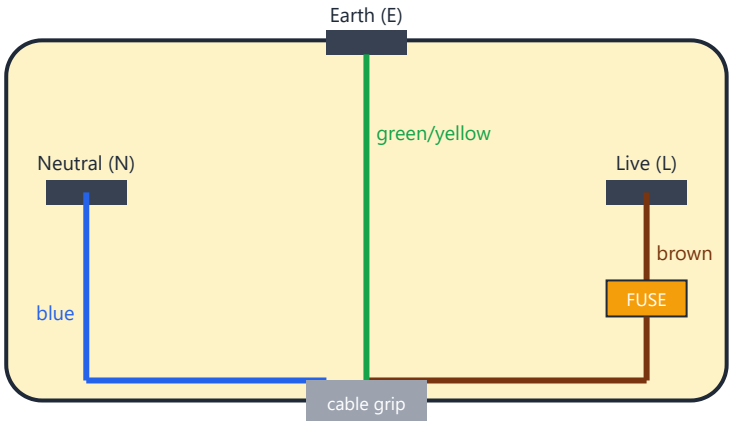
P3.7 Mains Electricity

UK mains electricity is **alternating current (a.c.)** at **230 V, 50 Hz**. Batteries supply **direct current (d.c.)** — the current flows one way.

Three-Pin Plug

- **Live (brown)**: at 230 V alternating. Dangerous.
- **Neutral (blue)**: at ≈ 0 V, completes the circuit.
- **Earth (green/yellow)**: safety wire. Normally carries no current; connects the metal casing of an appliance to the ground so that if the live wire touches the case, a large current flows to earth and blows the fuse.
- **Fuse**: thin wire that melts and breaks the circuit if the current exceeds its rating.

UK Three-Pin Plug Wiring



Why a.c. for the Grid?

- Transformers only work with a.c., allowing the grid to **step the voltage up** for transmission (minimising I^2R heat losses in cables) and **step it down** again at the user.

P3.8 Charge and Static

Insulators can be charged by **friction** — electrons are rubbed off one surface onto the other. The object that *gains* electrons is **negative**; the one that loses them is **positive**.

- Like charges **repel**; unlike charges **attract**.

- Sparks (e.g. shocks from a door handle) occur when the p.d. is high enough to ionise the air so charge jumps across.
- Static hazards: fuelling planes (grounding straps avoid sparks), volatile chemical factories.
- Static uses: photocopiers, spray painting, inkjet printers.

Exam Tip: Never say "positive charge flows". At GCSE, it is electrons (negative) that actually move; conventional current is drawn from + to − for historical reasons.

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